COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S4
DATE PERFORMED:	7/29/26	**DATE SUBMITTED:**	7/29/26
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ACTIVITY NO.4.1 [Array-Based vs. STL Stack Implementations]

1. PROCEDURE OUTPUT

What procedures did you perform?

Objective

- Implement a Stack Data Structure using two distinct approaches in C++: a custom Static Array implementation and the C++ Standard Template Library

Tasks & Intended Learning Outcomes (ILOs):

- **ILO A:** Implement a stack using the C++ STL (`std::stack`).
- **ILO B.1:** Implement a stack using a Static Array.

Submission Requirements:

- **Documentation:** Please use the official **Activity Template** for your write-up.
- **Source Code:** Submit your C++ file named as `LastName_Seatwork4_1.cpp`.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

```cpp
#include<iostream>
#include <stack>



    int main(){

        //push
        //pop
        //isfull
```

```
        //top
        //create an object:
        std::stack<char> stack1;

        std::cout<<"Is The stack empty? "<<stack1.empty() <<std::endl;
        stack1.push('j');
        std::cout<<"current top of The stack: "<<stack1.top()
<<std::endl;
        stack1.push('i');
        std::cout<<"current top of the stack:"<<stack1.top()<<std::endl;
        stack1.push('n');
        std::cout<<"current top of The stack: "<<stack1.top()
<<std::endl;
        std::cout<<"Is The stack empty? "<<stack1.empty() <<std::endl;
        std::cout<<"what is the size of the stack? "<<stack1.size()
<<std::endl;

        // emplace
        stack1.emplace('n');
        std::cout << "current top of the stack after emplace: " <<
stack1.top() << std::endl;

        // swap
        std::stack<char> stack2;
        stack1.push('n');
        stack1.push('i');

        stack1.swap(stack2);
        std::cout << "New top of stack1 after swap: " << stack1.top() <<
std::endl;
        std::cout << "New top of stack2 after swap: " << stack2.top() <<
std::endl;


        return 0;
    }
```

```cpp
#include <iostream>

// Constants & Global variables
constexpr int MAX_SIZE = 10;
int stackArr[MAX_SIZE];
int topIndex = -1; // Renamed to avoid clashing with function names

// Function prototypes
bool isEmpty();
bool isFull();
void top();
void push();
void pop();

int main() {
    int choice;
    bool running = true;

    // Menu loop so the user can perform multiple operations
    while (running) {
        std::cout << "\n============================\n";
        std::cout << " Stack Operations: \n";
        std::cout << " [1] Push\n [2] Pop\n [3] Is Empty?\n [4] Is
Full?\n [5] View Top\n [6] Exit\n";
        std::cout << "Enter your choice: ";
        std::cin >> choice;

        switch (choice) {
            case 1:
                push();
                break;
            case 2:
                pop();
                break;
            case 3:
                if (isEmpty()) {
```

```
                    std::cout << "The stack IS empty.\n";
                } else {
                    std::cout << "The stack is NOT empty.\n";
                }
                break;
            case 4:
                if (isFull()) {
                    std::cout << "The stack IS full.\n";
                } else {
                    std::cout << "The stack is NOT full.\n";
                }
                break;
            case 5:
                top();
                break;
            case 6:
                running = false;
                std::cout << "Exiting program...\n";
                break;
            default:
                std::cout << "Invalid choice! Please try again.\n";
                break;
        }
    }

    return 0;
}

// Function definitions:

bool isEmpty() {
    return (topIndex == -1);
}

bool isFull() {
    return (topIndex == MAX_SIZE - 1);
}
```

```
void top() {
    // Error catching: check if empty before accessing top element
    if (isEmpty()) {
        std::cout << "Stack Underflow! The stack is empty.\n";
        return;
    }

    // Prints the top element
    std::cout << "The top element is: " << stackArr[topIndex] <<
std::endl;
}

void push() {
    // Error catching: check if full before adding elements
    if (isFull()) {
        std::cout << "Stack Overflow! Cannot push more elements.\n";
        return;
    }

    int newData;
    // Ask the user for input
    std::cout << "Enter an integer to push: ";
    std::cin >> newData;

    // Increment top first, then assign the new data
    topIndex++;
    stackArr[topIndex] = newData;
    std::cout << "Successfully pushed " << newData << " to the
stack.\n";
}

void pop() {
    // Error catching: check if empty before popping elements
    if (isEmpty()) {
        std::cout << "Stack Underflow! Cannot pop from an empty
stack.\n";
        return;
    }
```

```
    // Display the item to be popped and decrement top
    std::cout << "Popping: " << stackArr[topIndex] << std::endl;
    topIndex--;
}
```

OBSERVATION:

The insertion function did not work because it only accepted the user's input but did not save it in the stack. The top variable increased, and the new value was not stored in stackArr. This was fixed by adding top++ and assigning the value using stackArr[top] = newData.Wrong C++ syntax and naming. Some keywords were written in uppercase, function names did not match because of different letter cases, and there were spelling mistakes such as imEmpty(). These issues were corrected by using the proper lowercase keywords, making the function names consistent, fixing the typos, and adding the missing variable declaration.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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